Yihong Chen 陈奕宏

Target Roles: Research Scientist — LLMs & AI Agents

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Employment

Postdoc 2025.07–2027.06 (expected)
Department of Electronic Engineering, Tsinghua University
Research: Large language models, GUI agents

Education

Ph.D. 2020.08–2025.06
Department of Mathematical Sciences, Tsinghua University
Research: Numerical methods for kinetic equations

B.S. 2016.08–2020.06
Department of Mathematical Sciences, Tsinghua University

Research Experience

Extreme Token Phenomena in Transformers — First Author — 2025–Present

- Investigated the mechanisms and causal relationship between attention sinks and massive activations in Transformers. Identified gradient sinks through backpropagation analysis, showing how attention sinks concentrate local gradients along the value pathway. Proposed that massive activations regulate local gradient pressure by RMSNorm in pre-norm Transformers. Designed V-scale to rescale gradients only along the value pathway during backpropagation. V-scale suppresses massive activations while preserving attention sinks, providing experimental support for the proposed mechanism. Paper accepted at NeurIPS 2026.

GUI Agent Evaluation and Memory Reuse — First Author / Coauthor — 2025–Present

- As first author, surveyed GUI agent benchmarks and developed a taxonomy covering GUI grounding and question answering, multi-step navigation, and open environments. Compared datasets, interaction environments, and evaluators. Published in TMLR (2026).
- As a coauthor, contributed to mobile agent research on executable experience graphs built from past interaction trajectories. Matching graphs to new tasks reduces step-by-step model calls in recurring workflows, aiming to improve efficiency and robustness to interface changes. Validated on HarmonyOS devices. Published in the IJCAI 2026 Demo track.

Modeling and Numerics of Kinetic Equations — First Author / Coauthor — 2021–2025

- Investigated moment methods for high-dimensional gas kinetic equations by developing discrete-velocity-direction models based on the minimum-entropy principle. Analyzed equilibrium states and entropy properties, reduced equilibrium computation to low-dimensional convex optimization, and numerically validated extensions of several one-dimensional methods to higher dimensions.
- Proposed a Poisson-kernel-based moment closure and a moment inversion algorithm for two-dimensional kinetic equations with fixed speed. Established well-definedness of the closure and convergence guarantees for the distribution approximation as the number of moments increases. Improved convergence speed and simplified closure computations in numerical implementations.

Project Experience

Rounding Effects and Training Stability in LLMs — Key Contributor — 2026.10–Present

- Huawei collaboration, currently in its initial phase. Investigates how low-precision arithmetic and rounding affect training dynamics, error propagation, loss convergence, and model performance, particularly training-inference mismatches during LLM reinforcement learning. Aims to inform the design of numerical methods and stable training through theoretical analysis and experiments.

Intelligent Cross-Device Services — Key Contributor — 2026.01–Present

- Contributed to the design of a cross-device workflow with Huawei to reduce branching complexity and maintenance costs: a powerful cloud model explores tasks, trajectories are converted into execution graphs, and a local model handles only intent recognition. Workflow reuse reduces repeated inference; validated on test-set tasks.
- Contributed to the design of joint quantization and structured pruning methods for language models on smartphones and other edge devices. The project deployed a small language model on-device and demonstrated cross-device sharing tasks, laying the groundwork for commercial deployment.
- Coordinated technical design, task execution, and delivery with the Huawei project manager, the faculty supervisor, and team members. Project outputs one submitted manuscript and two patent applications.

Industry-Specific Agent Algorithms — Key Contributor — 2025.11–Present

- Beijing Municipal Science and Technology Commission project (Z251100008125003). Coordinated implementation and validation of six algorithms for planning efficiency and execution reliability, including task-graph search, workflow optimization, intent recognition, tool calling, task execution, and reinforcement fine-tuning. Compiled results and documentation; coordinated project testing and reviews.

Publications

AI, Large Language Models, and Agents

[1] **Yihong Chen**, Zhouchen Lin, Quanming Yao. Attention sinks induce gradient sinks: Massive activations as gradient regulators in Transformers. *NeurIPS*, 2026. https://openreview.net/forum?id=c49EUKrp1G

[2] **Yihong Chen**, Shuai Wang, Yaqing Wang, Quanming Yao. A survey on benchmarks of LLM-based GUI agents. *TMLR*, 2026. https://openreview.net/forum?id=ri3yPWE21Q

[3] Xiaohan Zheng, **Yihong Chen**, Haiquan Qiu, Quanming Yao. vSpeedUI: Turning Past GUI Experience into Fast Executable Plans. *IJCAI Demo*, 2026.

[4] Ruibing Wang, **Yihong Chen**, Hongyi Nie, Shiqi Fan, Huixiang Zhang, Quanming Yao. EdgeQPS: Deployment-Aware Joint Quantization and Structured Pruning Search for Edge Language Models. 2026. (in submission)

[5] Yudong Bai, **Yihong Chen**, Quanming Yao, Yaqing Wang. DeltaReplay: Task-Relative Memory Reuse for Mobile GUI Agents. *arXiv:2610.11707*, 2026.

Modeling and Numerics of Kinetic Equations

[6] Ruixi Zhang, **Yihong Chen**, Qian Huang, Wen-An Yong. Dissipativeness of the Hyperbolic Quadrature Method of Moments for Kinetic Equations. *J Stat Phys*, 193(146), 2026. https://doi.org/10.1007/s10955-026-03706-y

[7] **Yihong Chen**, Qian Huang, Wen-An Yong, Ruixi Zhang. Poisson quadrature method of moments for 2D kinetic equations with velocity of constant magnitude. *Multiscale Modeling & Simulation*, 23(1), 2025. https://doi.org/10.1137/23m1620181

[8] **Yihong Chen**, Qian Huang, Wen-An Yong. Discrete-velocity-direction models of BGK-type with minimum entropy: II. Weighted models. *J Sci Comput*, 99(84), 2024. https://doi.org/10.1007/s10915-024-02531-3

[9] Qian Huang, **Yihong Chen**, Wen-An Yong. Discrete-velocity-direction models of BGK-type with minimum entropy: I. Basic Idea. *J Sci Comput*, 95(80), 2023. https://doi.org/10.1007/s10915-023-02211-8

[10] Tianshu Li, **Yihong Chen**, Qian Huang, Wen-An Yong. Combining hyperbolic quadrature method of moments and discrete-velocity-direction models for solving BGK-type equations. *arXiv:2411.12654*, 2024.

Honors and Awards

- 2025, Outstanding Doctoral Dissertation Award, Tsinghua University
- 2019, Comprehensive Excellence Award, Tsinghua University
- 2018, Academic Excellence Award, Tsinghua University
- 2017, Academic Excellence Award, Tsinghua University
- 2017–2020, Member of Tsinghua Xuetang Mathematics Program
- 2015, Second Prize, Chinese Mathematical Olympiad (CMO)

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